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Clinical Agreement in Wearable Technologies: A Bayesian Approach

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Abstract

This study assesses the ability of wearable technology to replace the clinical *Gold Standard* in a safe way by examining clinical equivalence rather than mere correlation between instruments. Using a Bayesian hierarchical approach (**brms** and INLA), we examined respiratory rate in patients with COPD and heart rate data across diverse conditions. The results indicate that while certain sensors, such as the *Apple Watch*, demonstrate high durability, others remain vulnerable to motion artifacts and physiological diversity. Consequently, many devices fail to meet the rigorous criteria established for clinical interchangeability through Bayesian methods. For instance, sensors such as *chest-bands* performed reliably during rest periods but failed to maintain clinical precision during physical exercise, highlighting the necessity of context-aware validation in medical wearables.

Keywords: Measurement Agreement, Bayesian Inference, Clinical Validation, COPD, Heart Rate.

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1. Introduction

In modern clinical practice, portable and wearable devices are increasingly being used instead of hospital-based equipment. This change is often supported by showing that these devices have a high correlation with an established reference standard. However, correlation alone is not enough to prove that one device can replace another in clinical care (Bland and Altman, 1990). A device may follow the same pattern as the reference method but still give readings that are consistently too high or too low. Correlation does not reveal this type of bias, and in clinical settings, it can lead to unreliable diagnostic decisions. For this reason, the key question is not whether two devices move in the same direction, but whether their measurements are close enough for one device to safely replace the other.

It should be clear from the start that this work is a descriptive agreement study, not a predictive one. The aim is not to predict the value of a signal, but to describe the difference between a portable device and its reference method, and to assess what that difference could mean clinically. Following the agreement framework proposed in Parker et al. (2020), this is assessed using measures such as bias, limits of agreement (LoA), Coverage Probability (CP), Total Deviation Index (TDI), and Concordance Correlation Coefficient (CCC).

Two clinical areas are analyzed, each with a defined reference standard. In the University of Edinburgh COPD respiratory rate dataset Parker et al. (2020), the reference standard is RRox, a respiratory rate measure derived from pulse oximetry and used as a clinical reference for patients with chronic respiratory diseases. In the heart rate dataset, the reference standard is the Bittium Faros 180, a medical-grade ECG patch recorded at around (~ 1000 Hz) and worn continuously throughout the protocol. Both datasets are described in more detail below. They are intentionally complementary: the COPD dataset is small, high-risk, and noisy, with limited structured data, while the heart rate dataset is larger and more diverse. This allows additional sources of variation to be examined, including activity type and skin tone based on the Fitzpatrick scale.

The contribution of this work is to extend the frequentist analysis Parker et al. (2020) into a Bayesian framework. While a frequentist approach provides point estimates and confidence intervals, a Bayesian model gives full posterior distributions (Campbell et al., 2023). This makes it possible to express clinical decisions in terms of probabilities. For example, a device may have an estimated Coverage Probability (CP) above 80%, but it may still be unsafe to use if the posterior probability of meeting that threshold is low. By modelling observations as nested within patients and activities, the analysis also accounts for both within-subject and between-subject variability. This helps avoid underestimating uncertainty, which can happen when all observations are simply pooled together.

2. Methods

This section describes the data sources, the preprocessing pipeline and the modelling. The two datasets are presented at the same time because, although they are different in size and context, they share a common analytical determination: a unified representation centred on the paired difference, which feeds a hierarchical Bayesian agreement model. All preprocessing was implemented in R and the models were fitted with *brms* and *INLA*. The deployment prototype was built in Python with *bayesflow* and *Streamlit*.

The unit of analysis is the paired difference:

$$\text{diff} = \text{Device Value} - \text{Reference Value} \quad (2.1)$$

where zero means perfect agreement, and any consistent difference shows the device’s pattern of error.

Following the SMART framework from the seminars, the descriptive goal is defined through a clear *Measurement of Success*. A device is recommended for clinical use only if its Coverage Probability (CP), which is the proportion of paired differences that fall within a clinically acceptable margin, meets the Bayesian decision rule: $P(\text{CP} \geq 80\%) \geq 70\%$. The acceptable margin is ± 6 rpm for respiratory rate and ± 5 bpm for heart rate.

The quality of the description is then evaluated using three criteria: (i) the posterior estimates of bias, LoA, CP, TDI and CCC and the width of their credible intervals; (ii) the satisfaction of the decision rule above, both globally and grouped by activity and skin tone; and (iii) the consistency of the final conclusion across datasets and alternative model specifications. Under this framework, the project does not only ask whether each wearable can replace its *Gold Standard*¹ in general, but also whether the wearable can be considered interchangeable under specific conditions.

2.1. Respiratory Rate in COPD Patients

The first dataset, released along [Parker et al. \(2020\)](#) by the University of Edinburgh, has 385 observations from 21 patients with chronic obstructive pulmonary disease. Each row records the respiratory rate (RR) measured simultaneously by six wearable channels: two cameras (*cm1*, *cm2*), two PPG sensors (*ppg1*, *ppg2*), an impedance band (*imp*), an accelerometer (*acc*) and a chest band (*cb*), together with the clinical reference *RRox*, derived from pulse oximetry and adopted as the Gold Standard throughout the study (Table 2.1). Patients perform a fixed sequence of activities (rest, talking, walking, recovery), so observations are nested within both patient and activity, a hierarchy that any valid agreement model has to respect.

¹In clinical research, a Gold Standard is the benchmark measurement regarded as the most accurate and reliable for a specific condition, against which new methods are compared.

Variable	Description	Type
Id/Time/Obs	Identification, time and observation number	Categorical /Time
Activity	Type of activity performed	Categorical
RRox	Gold Standard	Numerical
RR	Rate measured with a camera (cm1/2), PPG (ppg1/2), impedance (imp), accelerometer (acc) and chest band (cb)	Numerical

Table 2.1: COPD Variables.

The exploratory analysis (Figures A.1 and A.2) shows two structural problems that end up shaping the rest of the pipeline. First, the marginal distributions of RR are different between channels: RRppg1 is shifted upwards, RRcm1 is shifted downwards, and RRimp shows quartiles close to RRox but with tails that suggest instability. Second, although the RRox relationship is roughly linear, missingness is high 52% and it is clearly non random: certain devices simply cannot measure during certain activities. This is exactly the kind of situation where correlation becomes misleading, since its value depends on the subset that happens to be observed, and it is what motivated the agreement approach used here in the first place.

Preprocessing was therefore conservative. Identifier and activity variables were converted to factors so the hierarchical models could group them correctly. Non physiological readings with zero or negative values were recoded as missing. Instead of dropping variables, only the device–activity pairs with more than 20% missing data were excluded, which preserves as much clinical signal as possible while keeping biased estimates from observed combinations out of the analysis. Outliers within the $1.5 \cdot \text{IQR}$ range were kept on purpose: in chronic respiratory patients, sudden spikes during exertion are physiological events, not measurement noise, and throwing them out would artificially inflate the apparent agreement of the devices.

2.2. Heart Rate in Wearable Devices

The second dataset (Bent and Dunn (2021)) is used as an external validation of the framework on a much larger and more diverse population. It contains 238,595 heart rate records from 53 participants (with 32 F and 21 M, aged 18–54), distributed evenly across the six categories of the Fitzpatrick skin-tone scale, which is a dermatological classification that ranks skin types by their response to ultraviolet (UV) exposure, from Type I (very light, burns easily) to Type VI (very dark, rarely burns). This balanced design was on purpose: photoplethysmographic (PPG) sensors estimate heart rate from the optical absorption of green light by haemoglobin, and higher melanin levels attenuate that signal, so any agreement analysis of smart watches devices has to treat skin tone as a real source of variation rather than a nuisance covariate. Each participant followed a one-hour protocol divided into three rounds (Empatica E4 + Apple Watch 4; Fitbit Charge 2; Garmin Vivosmart 3 + Xiaomi Miband 3 + Biovotion Everion), with the Bittium Faros 180 ECG patch (1000 Hz) used as the Gold Standard across all

rounds. Each round followed the same activity sequence: seated rest for 4 min, paced deep breathing for 1 min, walking up to 50% HR max for 5 min, recovery for 2 min, typing for 1 min, so the design exposes every device to a comparable spectrum of motion artefacts. Variables are summarised in Table 2.2 and the full device specifications are reported in Table A.1.

Variable	Description (bpm)	Frequency/Variable
Faros 180ECG	Gold Standard	~ 1000 Hz
Watches	Apple Watch, Garmin, Fitbit, Miband	Variable
Pro Sensors	Empatica E4, Biovotion Ev-erion	~ 1 Hz
ID/SkinTone	ID ($n = 53$) and Fitzpatrick skin tone scale (1-6)	Categorical
Activity	Rest, Walking, Deep Breathing, Typing	Categorical

Table 2.2: Heart Rate Variables.

The exploratory analysis (Figures A.3 to A.7) tells a consistent story. The ECG is active throughout the protocol, while the smartwatches register less than 25% of the time (Figure A.3). The BPM range goes from a reasonable 40 BPM (athletic baseline) to an unreasonable 230 BPM, with all the extreme outliers concentrated in the ECG channel (Figure A.4). The chronological inspection of the protocol confirms that participants follow the prescribed Rest–Breathe–Activity–Type sequence (Figure A.5), but the volume of records is noticeably lower during rest and typing than during walking (Figure A.6) and the participant device coverage matrix is uneven, with Biovotion, Garmin and Xiaomi missing for many participants (Figure A.7).

This is the reason why the cleaning process had to deal with two different sources of noise: structural fragmentation (loss of skin contact, uneven coverage) and physiological improbability (ECG outliers). Rows with missing ECG were removed and incomplete device–reference pairs discarded, which reduced the dataset from 221,475 to 204,121 valid rows. Additionally, a physiological filter (30 to 200 BPM) was applied to the ECG channel to erase 91 residual artefacts, following the target heart rate guidelines established by the [American Heart Association](#) (sf). Since the heart rate dataset is two orders of magnitude larger than COPD, this stricter cleaning is affordable and provides credible intervals for the agreement metrics (CP, TDI, CCC) without sacrificing power.

2.3. Mineable View

After cleaning the datasets, both tables were transformed from a wide layout (one column per device) into a long format with one row per (subject, activity, device) measurement. The two datasets share the same analytical determination (one row, one paired comparison, one diff), but each retains the original column names produced by the pipeline. Table A.2 and Table A.3 document the schema of each mineable

view and the origin of every field. The design follows the Bland-Altman rationale extended by [Parker et al. \(2020\)](#), so the comparison between the Bayesian results and the published frequentist baseline is direct.

2.4. Model Building

All models share the same logic: the difference $diff_{ij}$ is modelled as a Normal distribution around a mean that is decomposed into a global bias α , a patient-level random effect $u_i \sim N(0, \sigma_{\text{patient}})$ capturing physiological and anatomical variability, and when applicable, an activity-level random effect $v_j \sim N(0, \sigma_{\text{activity}})$ absorbing the shift in error caused by motion. Treating the 385 COPD rows or the 204,121 heart-rate rows as independent would lead to overconfident inferences; the hierarchy propagates within and between-subject variability into every posterior quantity, including the final clinical decision. A device is recommended only if $P(CP \geq 80\%) \geq 70\%$, with the clinically acceptable margin set to ± 6 rpm for COPD and ± 5 BPM for heart rate.

2.4.1 Model Building COPD

Two specifications were fitted (Table 2.3). The *Global model* is fitted separately for each device against RRox and includes both random effects, providing one verdict per device. The *Chest-band model* is fitted one activity at a time and includes only the patient effect, isolating the impact of motion on the best-performing device identified globally.

Model	Specification	Description
Global	$diff_{ij} \sim \text{Normal}(\mu_{ij}, \sigma^2)$ $\mu_{ij} = \alpha + u_i + v_j$ $u_i \sim \text{Normal}(0, \sigma_{\text{patient}})$ $v_j \sim \text{Normal}(0, \sigma_{\text{activity}})$	Fitted separately per device against RRox. The outcome is $diff = \text{device} - \text{RRox}$.
Chest-band	$diff_{ij} \sim \text{Normal}(\mu_{ij}, \sigma)$ $\mu_{ij} = \alpha + u_i$	Activity-stratified (one model per activity) to assess within-activity agreement.

Table 2.3: Global and Chest-Band Models.

2.4.2 Model Building HEART RATE

Two complementary frameworks were used (Table 2.4). The *INLA stratified model* fits one model per (activity $\times$ skin-tone) subgroup, exploiting the speed of Integrated Nested Laplace Approximation to detect changes in bias driven by motion artefacts and melanin interference. The *brms global model* mirrors the COPD global specification on the full dataset, decomposing total error into patient and activity variance components and producing a single hierarchical verdict per device.

Model	Structure	Description
INLA	$\text{diff}_{ij} = \alpha + u_i$	Stratified: One model per subgroup to identify changes in bias based on Activity and Skin Tone.
brms	$\text{diff}_{ij} = \alpha + u_i + v_j + \epsilon$	Global: Hierarchical model that decomposes total error into patient and activity variance.

Table 2.4: INLA and BRMS Models.

2.4.3 Model Building Bayesflow

In order to overcome the computing inefficiencies associated with conventional MCMC and produce an easy-to-use tool for clinical application, Amortized Bayesian Inference was adopted using the *BayesFlow* library. While INLA or `brms` models rely on the computationally expensive sampling procedure, the *BayesFlow* model learns a neural network that provides almost instant inference [Radev et al. \(2022\)](#).

This approach is employed in relation to each data set, producing two distinct models adapted to specific characteristics of the respective clinical data sets and measurements. The evaluation of the *BayesFlow* model focuses on two main aspects: the stability of the amortized learning process and the precision of the resulting inference regarding key clinical parameters.

The learning process was measured using the Negative Log-Likelihood (NLL) loss function across 100 epochs. Both models exhibited a sharp initial drop followed by a smooth convergence towards an asymptote, indicating that the models effectively learned the posterior distribution of the data metrics.

Regarding the precision of the inferred clinical parameters, the Parameter Recovery plots (see Figures [A.8](#), [A.9](#), [A.10](#) and [A.11](#) in the Appendix) demonstrate very high precision for the bias and residual error parameters, while showing moderate precision regarding patient variability.

In conclusion, these results validate the use of *BayesFlow* as a robust inference engine. It provides agreement metrics with minimal estimation error and a processing speed that meets the requirements for live clinical decision support.

2.4.4 Model Evaluation

The modelling process was developed iteratively rather than as a single fixed analysis. We first used frequentist agreement metrics as a baseline, following [Parker et al. \(2020\)](#), and then moved to Bayesian hierarchical models to include uncertainty, patient-level variability and activity-level variability. For the COPD dataset, the first version was a global model fitted separately for each device. After seeing that the global verdict was hiding important differences between activities, we added an activity-stratified model for the chest-band, which was the best-performing device globally.

For the heart-rate dataset, the same logic was extended with two complementary approaches. INLA was used to fit stratified models by activity and skin tone, because

it allowed faster subgroup analysis on a larger dataset. A global brms model was also fitted to keep a comparable Bayesian hierarchical baseline across devices. Across all iterations, the evaluation protocol remained the same: bias, limits of agreement, Coverage Probability, TDI, CCC and CIA were used, and the final clinical decision was based on the posterior rule $P(CP \geq 80\%) \geq 70\%$.

Since this is a descriptive study rather than predictive, classical hyperparameter optimization was not the main objective. Instead, the improvement focused on model structure, stratification variables, Bayesian uncertainty and clinical decision rules. The repeated calculation of agreement metrics across devices and subgroups allowed the analysis to become more systematic and made the final interpretation more context-aware.

To ensure good clinical recommendations, the parameters were set using meaningful margins. For heart rate, a margin of ± 5 bpm was used, which is considered an industry benchmark for clinical accuracy according to the ANSI/AAMI EC13 standard aami2002. In the case of the COPD dataset, the margin is ± 6 rpm, which ensures that the physiological maxima in patients affected by the disease is captured with a minimal impact of measurement errors.

To measure success, a Coverage Probability (CP) of at least 80% was necessary, ensuring that the vast majority of readings fall within the safe ranges. However, because point estimates can be misleading in noisy or small datasets, the Bayesian decision rule requires at least a 70% posterior probability that the device meets the threshold $P(CP \geq 80\%) \geq 70\%$. This requirement helps reduce overconfidence, ensuring that a device is recommended only when there is high certainty about the reliability.

3. Results

This section reports the agreement metrics from the four models in Section 2.4 and turns them into a clinical verdict using the rule $P(CP \geq 80\%) \geq 70\%$. Frequentist and Bayesian estimates are reported next to each other on purpose, because the gap between the two is the most interesting thing the analysis showed. The full posterior tables (means, 95% credible intervals, posterior probabilities and variance components) are in Appendix A, and they are referenced inline.

3.1. Bayesian Metrics

The six metrics used (Table 3.1) are the same ones suggested by Parker et al. (2020), and each one was looked at twice: first as a frequentist point estimate, used as a baseline, and then as a posterior distribution from which a 95% credible interval and the posterior probability of meeting the clinical threshold were extracted. The recommendation is taken from that posterior probability, something that simply cannot be computed in the frequentist pipeline, which is the whole reason for moving to a Bayesian framework in the first place.

Detailed model results for the Global COPD, Stratified COPD, and Heart Rate models (INLA and BRMS) are provided in Table A.6 through Table A.13 within Appendix A.

Metric	Type	Definition & Clinical Threshold
Bias (α)	Systematic	Mean individual difference. A value near 0 indicates no systematic error.
Limits of Agreement (LoA)	Variability	The [95%] range of individual differences. Wide LoA signifies poor reliability.
Coverage Probability (CP)	Agreement	The % of errors within ± 6 rpm (COPD) or ± 5 BPM (HR). Threshold: $\geq 80\%$.
TDI	Risk	The maximum error expected in 95% of cases.
CCC	Precision	Combined measure of precision and accuracy (Threshold > 0.80).
CIA	Precision	Measures if the device adds more noise than the human variability itself.

Table 3.1: Frequentist decision metrics.

The Bayesian models also include a variance decomposition that splits the residual error into three sources: σ_{patient} (biological variability between subjects), σ_{activity} (induced by motion variability between tasks) and σ_{residual} (sensor noise). This is what

later allows the verdict to be conditional on activity and skin tone, instead of a single global number that hides everything underneath.

3.2. COPD Results: Global Model and Chest Band Model

In the global COPD analysis, no device meets the Bayesian threshold (Table A.6 and A.7). The Chest-band gets the highest agreement of all the devices ($CCC = 0.717$, $TDI = 10.1$ rpm, Table A.7), which means that, even for the best device, in 95% of measurements the error is expected to be up to 10 rpm away from the oximetry reading; this is almost twice the ± 6 rpm clinical margin. The accelerometer comes second ($CCC = 0.596$, Table A.7), and the camera and PPG devices are basically unusable because the missing data is too much and not random (52%), so the credible intervals come out very wide (Table A.6). The camera model, on top of that, throws 88 divergent transitions, which is a clear sign that the MCMC is unstable because there is just not enough data. The CIA values for these devices go above 1 (Table A.7), meaning the sensor adds more noise than the variability between patients; at that point, the device is not really measuring anything useful, but merely adding error on top of the natural differences between people.

The Chest-band is the only device that a frequentist framework would have approved (Table A.6 and A.7). Its frequentist CP is 80.8%, just above the 80% threshold, which would mean that the device stays within ± 6 rpm of RRox in 8 out of 10 readings. But the Bayesian model puts the mean CP at 72.1% and gives only $P(CP \geq 80\%) = 1.5\%$ (Table A.6), which means there is only a 1.5% chance that the device truly meets the 80% safety bar, and a 98.5% chance that it does not. What is happening is simple: the frequentist point estimate is sitting in the upper tail of a posterior distribution that is mostly below the threshold, so the apparent "pass" is really an issue of ignoring the uncertainty in a small dataset with hierarchical structure. In practice, trusting the frequentist verdict here would have meant putting a device into clinical use with a 98% probability of failing the safety criterion.

The activity-stratified Chest-band model brings the answer that the global model was hiding (Table A.8 and A.9). During activities with low movement, the posterior CP goes up to almost 1 and the device is *RECOMMENDED*, meaning that under those conditions every reading falls inside the ± 6 rpm clinical margin. During exercise, the posterior CP basically falls to 0%, which is the opposite extreme: almost no reading stays inside the safe range when the patient is moving. The variance decomposition (Table A.9) makes the reason obvious: σ_{activity} dominates σ_{patient} , so the global average is being pulled down by motion errors rather than by differences between patients. Therefore, the accurate conclusion is not that the chest-band can replace RRox, but that the chest-band can replace RRox under low-movement conditions, and only there, which is exactly the kind of conditional recommendation that motivated this project.

3.3. SMARTWATCHES Results: INLA Model and BRMS Model

The heart-rate analysis was run with two complementary inference ways. The INLA stratified model fits one cell at a time (Table A.10 to A.12) and the brms global model gives one hierarchical verdict per device (Table A.13). Both end up agreeing on the same thing, that no device accomplishes the Bayesian threshold in clinical terms, none of the smartwatches tested can be safely used as a replacement for an ECG patch under the criteria adopted in this work.

The INLA results show a lot of heterogeneity. The Apple Watch is the most consistent of all, with CP between 0.35 and 0.45 in the Fitzpatrick scale (Table A.12) and only a little fluctuation in the activities (Table A.11), which means that only 35 to 45% of the readings stay in the ± 5 BPM margin of the ECG, depending on the skin tone of the person, this is far below the 80% bar. Biovotion and Empatica fall apart during everyday tasks, with CP values close to zero (Table A.10), meaning almost none of their readings are in the clinical range while the user is moving. Garmin shows an error of -119.64 BPM (Table A.10), which is too big to be real and looks more like a synchronisation problem in the export than an actual sensor failure, on average, Garmin reports a heart rate 120 BPM lower than the ECG, an error of that size can't come from the sensor and certainly reflects misaligned timestamps. The CP and CCC distributions broken down by Fitzpatrick type (Figures A.12 and A.13) confirm that motion noise and melanin interference act as non random error triggers, agreement gets worse as skin tone darkens, so the same device that works well on lighter skin (Type I and II) becomes unreliable on darker skin (Type V and VI), which is exactly what would be expected from the optical attenuation that PPG suffers with higher melanin levels.

The brms model (Table A.13) confirms the same under a single hierarchical specification. Even the best device, Apple Watch, ends up with a global posterior CP of 0.005 and a TDI = 39.85 BPM (Table A.13), meaning only 0.5% of readings are in the ± 5 BPM clinical range, and in 95% of cases the error can be up to 40 BPM, almost eight times the safe range. Garmin's CIA of 1 (Table A.13) again indicates a data issue rather than the sensor failing. The variance decomposition (Table A.13) returns $\sigma_{\text{activity}} > \sigma_{\text{patient}}$, the same pattern as COPD: that motion is the principal cause of disagreement, and the skin tone accentuates it.

3.4. BAYESFLOW Results: Deployment Mockup

Bayesflow models do not accomplish the same precision as the brms and INLA methods (detailed in Table A.7 for the COPD dataset and Table A.13 for the Smartwatches dataset). However, the resulting agreement metrics remain consistent across the evaluated devices. For the COPD data (Table A.4) and the Smartwatches data (Table A.5), the estimated TDI and CCC values follow the same clinical trends as the classical models, successfully identifying the same limitations in device substitutability.

The primary advantage of this approach lies in the transition from costly sam-

pling to a more efficient framework. By utilizing a pre-trained neural network, the system bypasses the significant time and computational overhead required by traditional Bayesian methods. This allows for near-instant evaluations, making the *BayesFlow* model a more suitable engine for a real-time production environment while maintaining a robust balance with statistical rigor.

The *BayesFlow* models do not accomplish the same results as the `brms` or INLA, but the agreement evaluation is still similar enough, and the computational and time cost upgrade allows for real-time clinical responses.

A prototype (Table A.14 and Table A.15) has been developed using `Streamlit` to automate concordance analysis. This interface is designed for descriptive agreement analysis using the *BayesFlow* model, not a predictive model. The user uploads a dataset, selects the Gold Standard column and the device column, chooses the device type only to apply the appropriate clinical tolerance, and receives agreement metrics with an interpretable clinical verdict instantly. This verdict translates the statistical results (like CP and TDI) into a simple, actionable phrase for the user.

4. Discussion

4.1. Beneficiaries, Limitations & Risks

The main value of the project lies not only in classifying the different devices, but also in providing a clinical decision-making framework that enables us to determine when a device may be interchangeable with its standard counterpart. As demonstrated by the COPD case, the final decision is influenced by the context and specific conditions. The novelty of this project lies in the strategic transition from traditional frequentist point-estimates to a Bayesian hierarchical framework for clinical agreement. Rather than producing an overall verdict that ignores statistical uncertainty, the model takes into account patient and environmental variability to generate accurate recommendations. This shift toward personalised safety metrics aligns with modern regulatory trends, such as those advocated by the **FDA** regarding the use of Bayesian statistics to better manage clinical risk and variability in medical device development ([U.S. Food and Drug Administration, 2010](#)).

The main beneficiaries are medical experts and device developers. The analysis generated helps to illustrate its real-world applicability: medical experts could use the website to upload pilot data from a low-cost commercial wearable device and use it to help determine whether it is safe to use for the continuous monitoring of hospitalised patients, reaching a clinical verdict without having to interpret external statistics. Furthermore, developers can identify the source of measurement errors, such as determining in which profiles (such as skin tones) or activities their sensors lose accuracy. It can also help explain to users when the device is not recommended.

The expected impact of this project is to speed up the validation of new medical devices. However, there are associated risks, the main one being over-reliance on the model's verdict as a final medical decision. The model should be used as a support tool and must never replace clinical judgement. There is also a risk of over-generalising the results if it is assumed that a device validated under certain conditions, such as low movement, is valid in all circumstances.

On the other hand, there are limitations due to the quality and nature of the original data; for example, non-random missing data in the COPD dataset, which limits the reliability of some estimates and makes it difficult to analyse certain combinations of device and activity. Furthermore, there is variability in the heart rate dataset. Another limitation is the source of the data, which comes from a controlled experiment, thereby limiting its direct applicability to real-world populations. Finally, the acceptance threshold set is static, whereas in some real-world scenarios where there is a higher diagnostic risk, stricter values may be required.

4.2. Legacy, Dissemination & SDGs

This project is based on reproducibility and open science. To ensure that the scientific and clinical community can audit or replicate the study, all the assets generated have been organised into a public repository on GitHub [De Paula Gonçalves et al. \(2026\)](#)¹. This repository contains the *R* scripts for data pre-processing, the Bayesian models and the *Python* code for the *Streamlit* website. The aim of this approach is to enable future teams to test new datasets or devices without having to start from scratch. Another way of disseminating the project is to share it in the form of a short report for other students, using the work of [Parker et al. \(2020\)](#) as a starting point.

The project has shown that a high correlation does not imply clinical agreement; correlation alone is not sufficient to determine whether a portable device can replace a clinical reference. Furthermore, uncertainty must be taken into account in the final decision, as a device that would otherwise be acceptable may not be so if its posterior probability of reaching the threshold is low. Finally, the creation of the website demonstrated that a statistical model is only truly valuable if its results are interpretable and applicable to the end user.

The growth plan for this project is based on expanding the current website prototype to automate data quality reports and serve as a useful tool; at an academic level, the aim is to publish a scientific article based on the methodology used, as well as to promote the repository as educational material.

In regards to Sustainable Development Goals, the project is connected to SDG 3 (Good Health and Well-being), addressing targets 3.4 and 3.d, reducing premature mortality from non-communicable diseases such as COPD and cardiovascular conditions through safer continuous monitoring and strengthening health risk management capacity by providing public systems with a transparent way to decide when low-cost wearables are safe substitutes for clinical equipment. The project also contributes to SDG 9 (Industry, Innovation and Infrastructure), specifically target 9.4, by modernising the methodological infrastructure for the validation of medical devices through the use of post-hoc distributions, and by creating a *Streamlit* prototype that is accessible to users without rebuilding the analytical pipeline from scratch ([United Nations, 2015](#)).

4.3. Use of technology

Throughout this project, *RStudio* served as the main environment for development due to its statistical specificity. For data processing and visualization, the **tidyverse** suite of packages was utilized, specifically **dplyr** and **tidyr** for data manipulation purposes, and **ggplot2** for generating clinical boxplots, agreement matrices, and other diagnostic graphs presented in this report.

Regarding the Bayesian approach, two different methods were employed. To create generalized linear mixed models within the Bayesian framework, the **brms** package was chosen due to its flexibility in model construction. To handle the large Heart Rate dataset, the INLA (*Integrated Nested Laplace Approximation*) method was selected as

¹<https://github.com/Miicchh/Agreement>

a computationally efficient alternative to traditional MCMC sampling. Additionally, Amortized Bayesian Inference using the *BayesFlow* library was implemented through *Python*, leveraging neural networks for real-time posterior estimation.

Finally, the report was authored in L^AT_EX using *Overleaf*, and the project’s operational mockup was developed as a web application using the *Python* library **Streamlit**. This application facilitates the translation of complex statistical results into a visual data product, allowing users to upload datasets, select the Gold Standard and device columns, and obtain an interpretable, actionable agreement verdict.

4.4. Seminar and AI use

Various concepts from the seminars and support tools such as Artificial Intelligence (AI) were used to develop the project. Firstly, the distinction between clinical objectives (replacing the Gold Standard) and data analysis objectives (measuring agreement between devices) was used to define a clear Success Metric, whereby a device is only recommended if $P(CP \geq 80\%) \geq 70\%$, using ± 6 rpm for respiratory rate and ± 5 bpm for heart rate.

Moreover, in the task taxonomy for Unit 3, the problem was framed as a descriptive analysis rather than a predictive one, as the aim is to assess whether a device is interchangeable, not to predict future values. The task is therefore framed as descriptive, rather than predictive or classificatory. This justified the use of more specific agreement metrics (bias, LoA, CP, TDI) rather than traditional regression metrics ($RMSE$ or R^2), as correlation does not imply agreement [Parker et al. \(2020\)](#).

The concept of iterative improvement was also applied: global models were initially constructed, but stratified analyses by activity and skin tone were then added when it became apparent that the global results masked significant differences. This was particularly useful for the chest band, which was not replaceable at a global level but became replaceable in conditions of low mobility.

Finally, AI tools (*ChatGPT*, *Gemini* and *Perplexity*) were used to structure the document and the code. However, limitations were identified during their use. For example, the AI suggested removing all null values from the COPD dataset, but missing values were not random, as some devices failed more often in specific activities, so it would have skewed the study. Furthermore, the tools simplified the conclusions to an overall verdict per device, ignoring the fact that reliability can change under certain conditions. For this reason, AI was used solely as a tool to assist and support the process, with the team reviewing and adapting all the results.

5. Conclusions

This project demonstrates that clinical agreement requires a probabilistic and contextualised approach, rather than a simple yes or no result. By applying the framework of [Parker et al. \(2020\)](#), within a Bayesian hierarchical model, certain deficiencies in classical frequentist criteria are observed. In the COPD results, the chest band appears acceptable under the frequentist criterion, with a PPV of 80.8%, but the Bayesian model estimates a posterior PPV of 72.1% and only a 1.5% probability of actually exceeding the 80% threshold. However, the Bayesian model also avoids the opposite error of completely rejecting the device. The model incorporating activity analysis shows that the chest strap can be reliable under conditions of low mobility, whilst its accuracy deteriorates during more intense movements. Its reliability therefore depends largely on the clinical context in which it is used.

The heart rate dataset confirms the same finding in a broader context, but with a more stricter result. The best-performing device, the Apple Watch, fails to meet clinical standards across all skin tones (CP of 0.35–0.45; TDI of ± 40 bpm compared to a safe margin of ± 5 bpm). In other words, although they are useful for day-to-day monitoring and outperform their competitors, current smartwatches cannot reliably replace a medical-grade ECG patch.

Ultimately, the results show that current wearable devices alone are insufficient to meet the standards required for full clinical replacement. However, this limitation is also one of the most important findings of the project, identifying their limits and the specific conditions under which they become useful. Overall, the main contribution of this work is the framework itself: a Bayesian agreement pipeline that turns device evaluation into a probabilistic and context-aware decision.

6. References

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A. Appendix

A.1. Exploratory Data Analysis

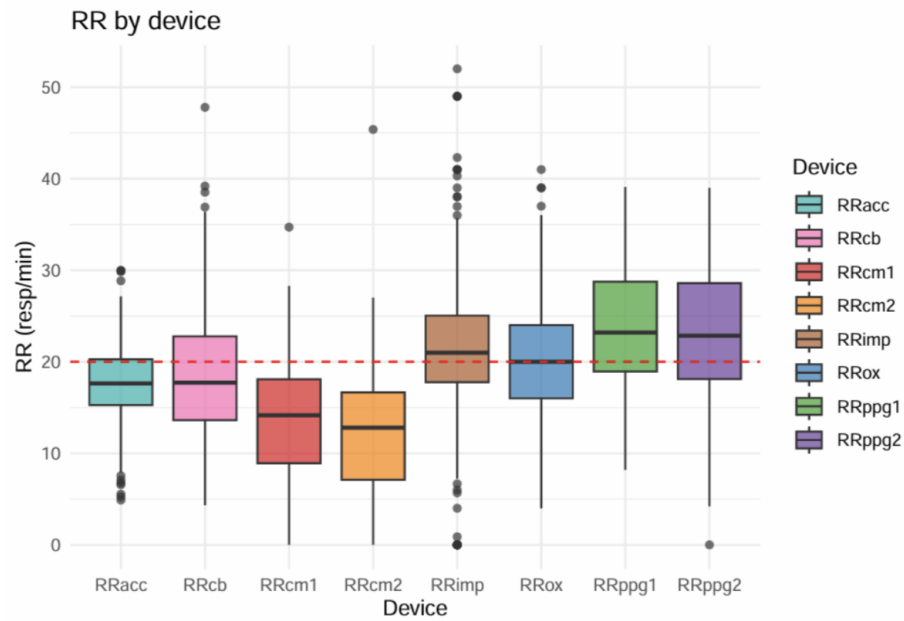


Figure A.1: Boxplots RR by Device

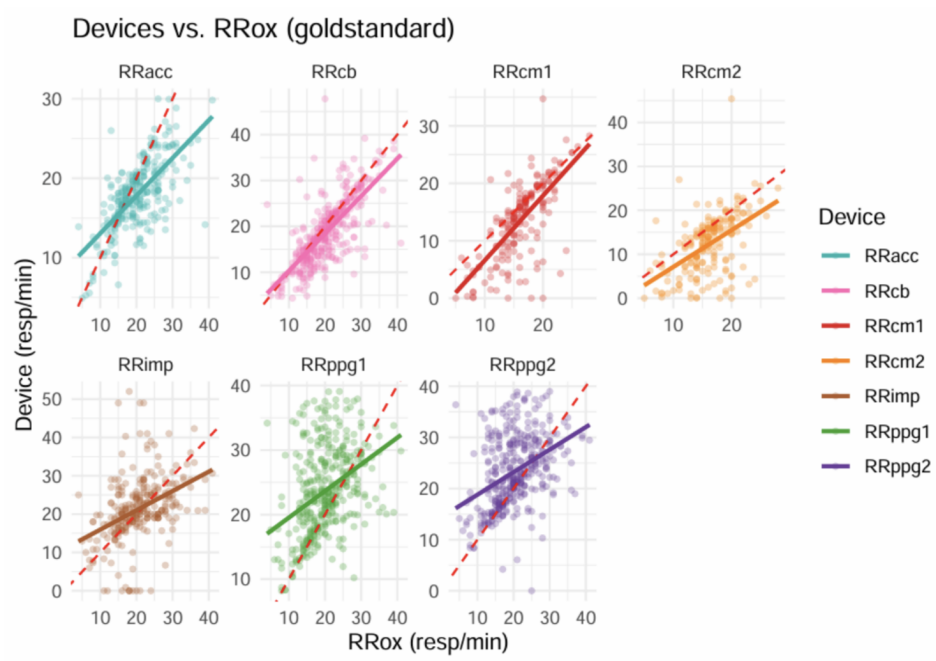


Figure A.2: Devices vs GoldStandard

Device	Manufacturer	Location	Release	Software
Apple Watch 4	Apple Inc.	Cupertino, CA	Fall 2018	5.1.3
Fitbit Charge 2	Fitbit, Inc.	San Francisco, CA	Fall 2016	22.55.2
Garmin Vivosmart 3	Garmin Ltd.	Olathe, KS	Spring 2017	5.10
Xiaomi Miband 3	Xiaomi Corp.	Beijing, China	Spring 2018	NA
Empatica E4	Empatica Inc.	Milano, Italy	NA	Summer 2019
Biovotion Everion	Biovotion AG	Zurich, Switzerland	NA	Summer 2019
Bittium Faros 180 (GS)	Bittium Corp.	Oulu, Finland	—	—

Table A.1: Device Specifications for the heart rate dataset

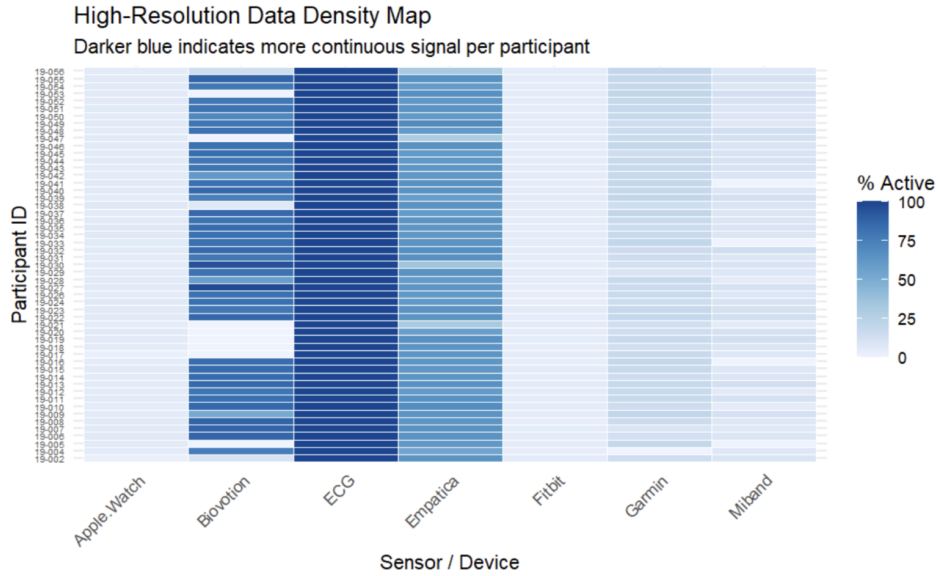


Figure A.3: Sensor Activity Profile per Participant

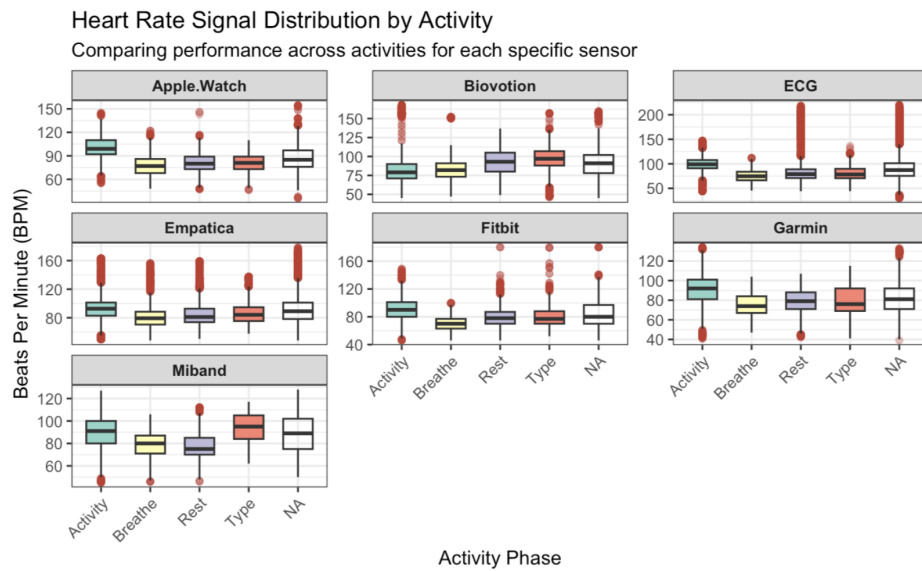


Figure A.4: Heart Rate Distribution by Activity and Device

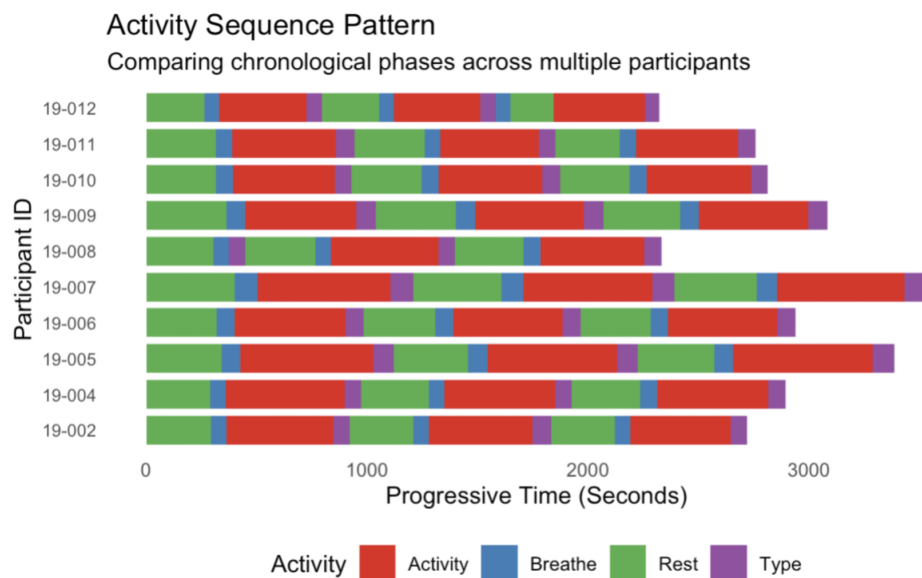


Figure A.5: Activity Sequence Pattern

Column	Type	Origin	Role
PatientID	Categorical (factor)	Original PatientID	Grouping variable for patient random effect u_i .
Obs	Integer	Original observation index	Preserves measurement order; not used as predictor.
Activity	Categorical (factor)	Original Activity column (rest, talking, walking, recovery)	Grouping variable for activity random effect v_j .
Device	Categorical (factor)	pivot_longer (names_to = "Device")	Identifies the wearable; one model fitted per device.
RR	Numerical (rpm)	pivot_longer (values_to = "RR")	Device measurement paired with RRox. ≤ 0 set to NA; device-activity pairs with $> 20\%$ missing excluded.
RRox	Numerical (rpm)	Original RRox	Gold Standard reference value.
diff	Numerical (rpm)	$RR - RRox$	Outcome variable of Bayesian models.

Table A.2: Mineable view for COPD dataset

Column	Type	Origin	Role
ID	Categorical (factor)	Original participant identifier (n = 53)	Grouping variable for patient random effect u_i .
Skin.Tone	Categorical (Fitzpatrick I–VI)	Original Skin.Tone column	Stratification variable; key for PPG analysis.
Activity	Categorical (factor)	Original Activity column (rest, deep breathing, walking, typing)	Grouping variable for activity random effect v_j .
Device	Categorical (factor)	pivot_longer (names.to = "Device")	Identifies which wearable produced each row; one model is fitted per device.
HR_Device	Numerical (bpm)	pivot_longer	Device measurement (bpm).
HR_GoldStandard	Numerical (bpm)	Original ECG	Reference value (30–200 BPM filter).
diff	Numerical (bpm)	$HR_{Dev} - HR_{GS}$	Outcome variable of Bayesian models.

Table A.3: Mineable view for HEART RATE dataset

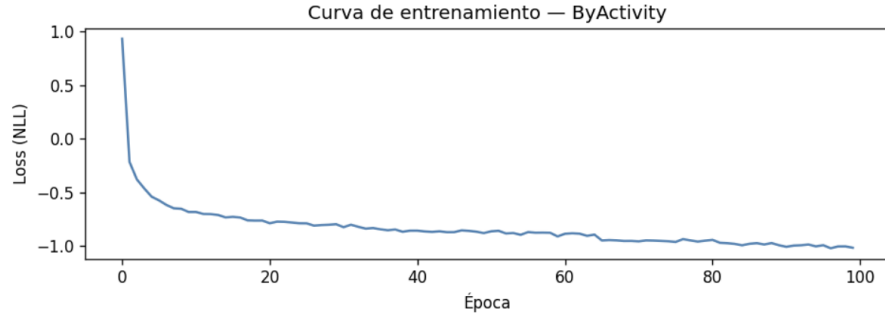


Figure A.8: COPD bayesflow Recovery Plot

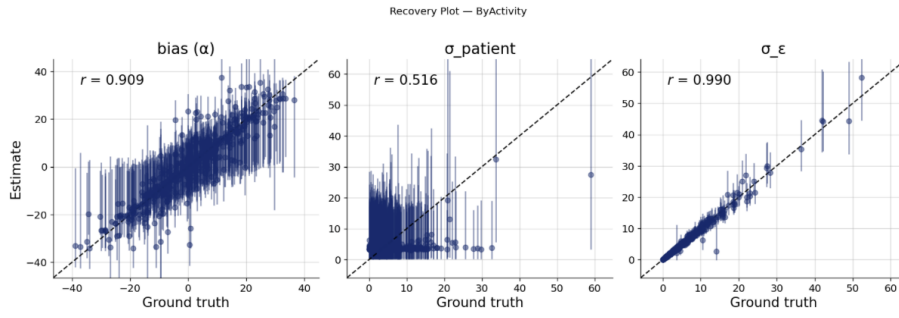


Figure A.9: COPD bayesflow Recovery Plot

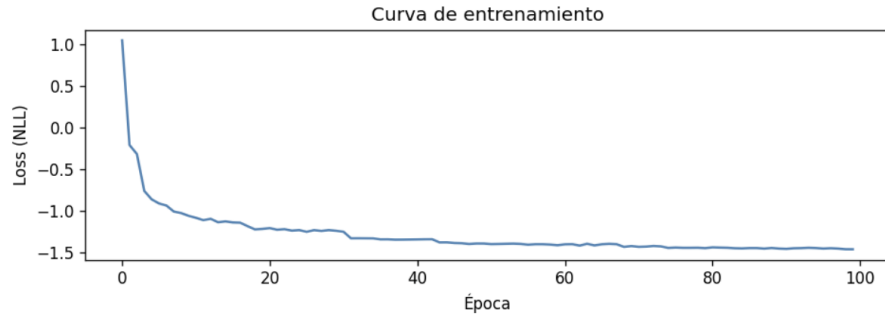


Figure A.10: Smartwatches bayesflow Recovery Plot

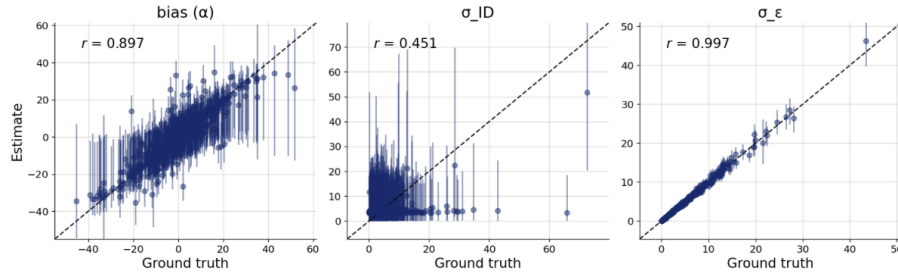


Figure A.11: Smartwatches bayesflow Recovery Plot

Device	TDI	CCC	CIA	Verdict
Camera 1	25.63	0.594	0.231	Not substitutable
Camera 2	28.30	0.379	0.252	Not substitutable
PPG 1	27.18	0.339	0.252	Not substitutable
PPG 2	25.75	0.354	0.245	Not substitutable
Impedance	23.95	0.231	-0.927	Not substitutable

Table A.4: Global Bayesian Agreement: TDI, CCC and Clinical Verdict (COPD BayesFlow model)

Device	CCC	LoA Inf	LoA Sup	TDI (95%)	CP (5 BPM)	CIA
Apple Watch	0.852	-25.018	36.678	35.720	0.239	0.110
Empatica	0.337	-17.177	13.701	19.393	0.459	0.348
Fitbit	0.503	-28.335	23.291	27.405	0.287	0.148
Biovotion	0.210	-10.333	33.148	31.654	0.183	0.210
Garmin	0.593	-23.447	28.159	28.895	0.285	0.153
Miband	0.349	-36.393	21.693	33.997	0.229	0.134

Table A.5: Global Bayesian Agreement (BayesFlow): TDI, CCC and Clinical Verdict (Smartwatches Model)

A.2. COPD Agreement Results

Device	Bias (rpm)	LoA (rpm)	CP_freq	CP_bayes (95%)	P(CP $\geq$ 80%)
Camera 1	-3.29	[-14.3, 7.7]	77.2%	63.1% [38.8%, 75.7%]	0%
Camera 2	-5.83	[-21.1, 9.5]	69.0%	42.9% [20.2%, 60.7%]	0%
PPG 1	2.86	[-11.9, 17.6]	62.4%	53.9% [46.4%, 59.9%]	0%
PPG 2	2.54	[-12.7, 17.8]	63.2%	53.2% [45.7%, 59.0%]	0%
Impedance	-4.57	[-27.4, 18.2]	55.2%	36.4% [27.9%, 42.9%]	0%
Accelerometer	-3.46	[-13.9, 7.0]	82.2%	63.6% [42.7%, 78.6%]	1.3%
Chest-band	-1.27	[-11.7, 9.2]	80.8%	72.1% [60.3%, 79.4%]	1.5%

Table A.6: Global Bayesian Agreement: Bias, LoA and Coverage Probability

Device	TDI	CCC	CIA	Verdict
Camera 1	12.6	0.561	0.132	Not substitutable
Camera 2	18.9	0.363	-0.457	Not substitutable
PPG 1	15.2	0.338	0.19	Not substitutable
PPG 2	15.4	0.353	0.199	Not substitutable
Impedance	23.8	0.234	-0.927	Not substitutable
Accelerometer	12.3	0.596	0.678	Not substitutable
Chest-band	10.1	0.717	0.678	Not substitutable

Table A.7: Global Bayesian Agreement: TDI, CCC and Clinical Verdict

Activity	Bias(rpm)	LoA(rpm)	CP_Freq	CP_bayes	P(CP $\geq$ 80%)
Standing	-1.8	[-5.6, 2.0]	100%	98.1%	100%
T Flat	-0.85	[-3.8, 2.1]	100%	99.8%	100%
TS 4%	0.04	[-2.9, 3.0]	100%	99.9%	100%
Sitting	-1.43	[-7.7, 4.9]	89.7%	90.9%	99.8%
Lying	0.47	[-4.8, 5.8]	90.5%	96.2%	99.7%
Sweeping	1.72	[-8.4, 11.8]	66.7%	72.5%	15.9%
Fast 6mWT	1.34	[-12.5, 15.2]	81%	59.2%	0.2%
Slow 6mWT	2.14	[-11.9, 16.2]	81%	57.4%	0.1%
Climbing S	-6.57	[-20.5, 7.4]	52.4%	42.4%	0%
Lifting	-4.98	[-15.6, 5.6]	52.4%	55.0%	0%
Stnd&W	-3.82	[-14.0, 6.4]	62.7%	62.9%	0%

Table A.8: Activity-Stratified Agreement: Bias, LoA and CP

Activity	TDI	CCC	CIA	Verdict
Standing	4.96	0.863	0.855	RECOMMENDED
T Flat	3.29	0.965	0.964	RECOMMENDED
TS 4%	2.75	0.968	0.967	RECOMMENDED
Sitting	6.71	0.719	0.651	RECOMMENDED
Lying	5.07	0.74	0.749	RECOMMENDED
Sweeping	10.2	0.613	0.371	NO
Fast 6mWT	13.3	0.445	0.572	NO
Slow 6mWT	14	0.32	-0.19	NO
Climbing S	18.3	0.118	-0.432	NO
Lifting	13.9	0.381	0.106	NO
Stnd&W	12.4	0.474	0.24	NO

Table A.9: Activity-Stratified Agreement: TDI, CCC and Verdict

A.3. Heart Rate Agreement Analysis

Device	Bias (α)	TDI (95%)	CP (5 BPM)	CIA	CCC
Apple Watch	-0.17	18.52	0.405	0.05	0.85
Fitbit	-4.20	34.54	0.228	0.14	0.50
Miband	-2.44	37.43	0.209	0.23	0.35
Empatica	-0.19	38.68	0.198	0.22	0.34
Biovotion	-3.99	46.73	0.164	0.16	0.21
Garmin	-119.64	1000	0.000	1.00	0.59

Table A.10: Global Bayesian Agreement (INLA)

Device	Rest (CP)	Breathe (CP)	Activity (CP)	Type (CP)
Apple Watch	0.43	0.36	0.54	0.59
Fitbit	0.30	0.45	0.22	0.25
Miband	0.25	0.35	0.00	0.16
Garmin	0.00	0.47	0.00	0.33
Empatica	0.23	0.00	0.21	0.00
Biovotion	0.20	0.00	0.00	0.00

Table A.11: Global Bayesian Agreement (INLA) by Activity

Device	Tone 1	Tone 2	Tone 3	Tone 4	Tone 5	Tone 6
Apple Watch	0.42	0.41	0.35	0.45	0.41	0.40
Fitbit	0.00	0.27	0.22	0.20	0.23	0.21
Miband	0.23	0.24	0.21	0.18	0.21	0.23
Empatica	0.23	0.22	0.19	0.18	0.20	0.21
Garmin	0.00	0.00	0.27	0.21	0.00	0.27
Biovotion	0.20	0.19	0.18	0.18	0.16	0.13

Table A.12: CP by Skin Tone (Fitzpatrick Scale)

Device	CCC	LoA Inf	LoA Sup	TDI (95%)	CP (5 BPM)	CIA
Apple Watch	0.123	9.893	42.647	39.848	0.00575	0.160
Empatica	0.022	38.052	92.692	87.923	0.00013	0.120
Fitbit	0.061	21.094	73.902	70.053	0.00075	0.144
Biovotion	-0.005	87.639	131.245	128.278	0.00000	0.725
Garmin	0.053	21.915	61.652	58.175	0.00013	0.311
Miband	0.013	43.501	91.279	87.056	0.00000	0.231

Table A.13: Global Bayesian Agreement (brms)

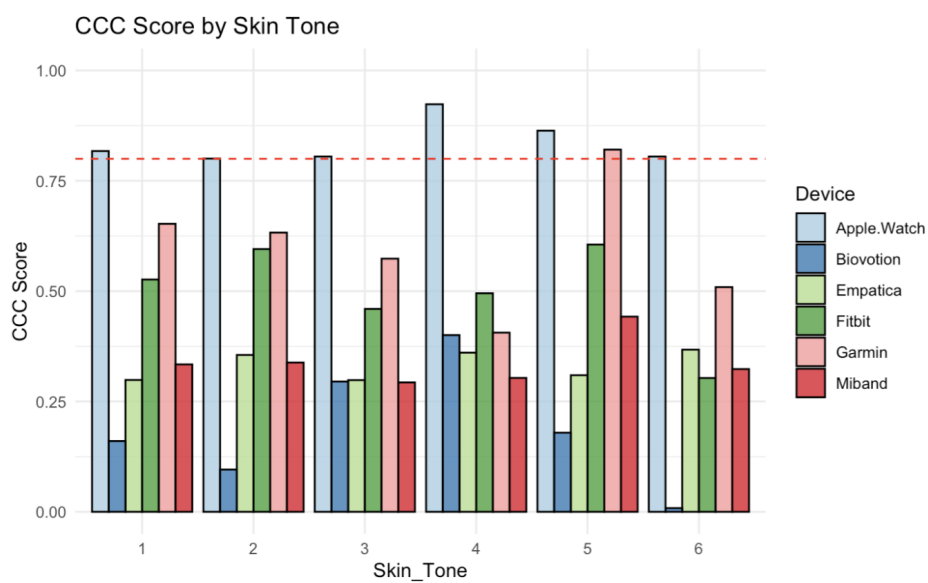


Figure A.12: CCC Score for device by Skin Tone

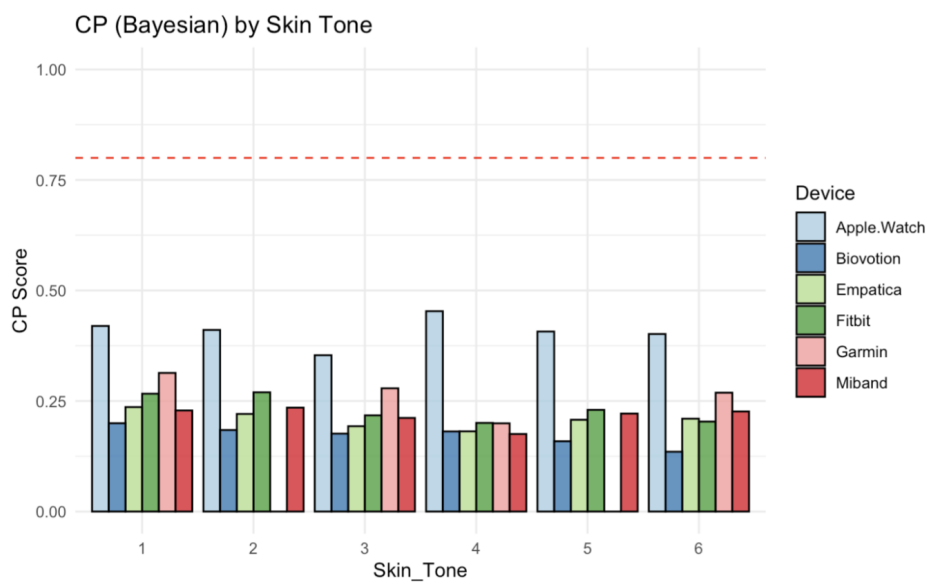


Figure A.13: CP Score for device by Skin Tone

A.4. Deployment Mockup

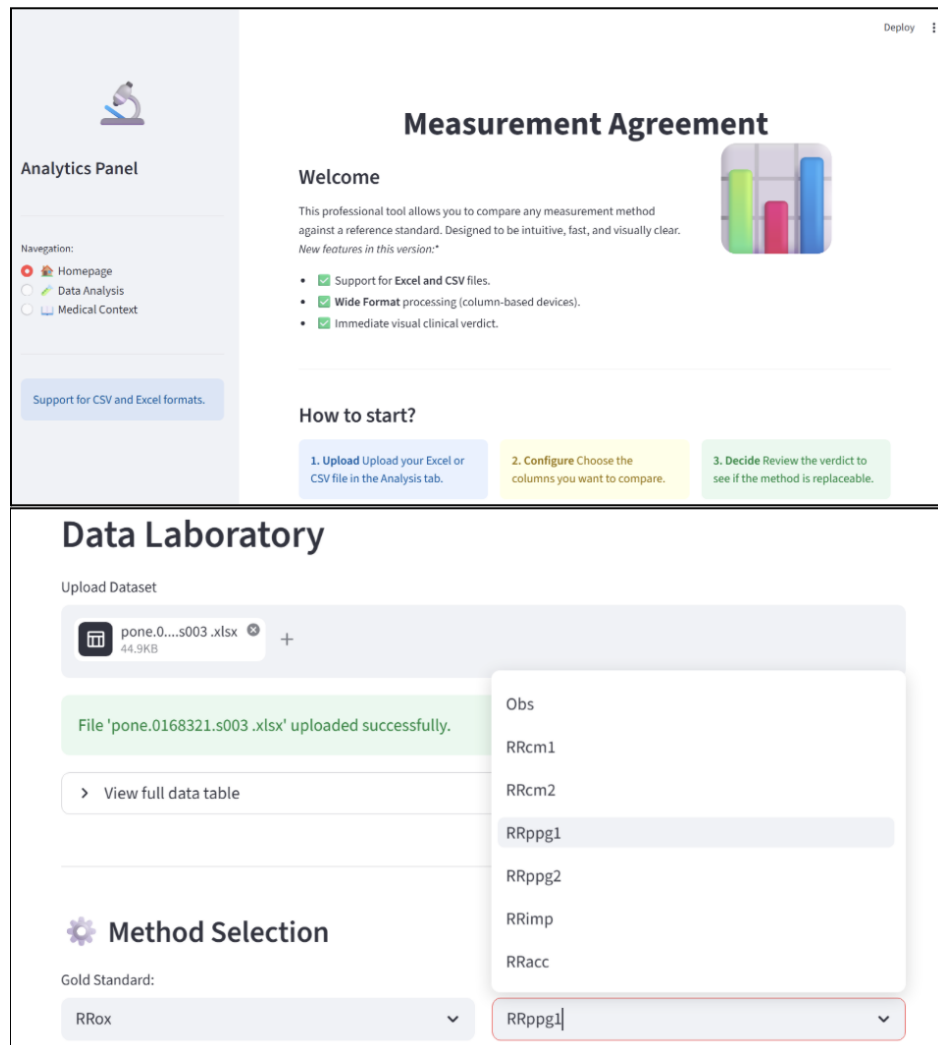


Figure A.14: Mockup Interface

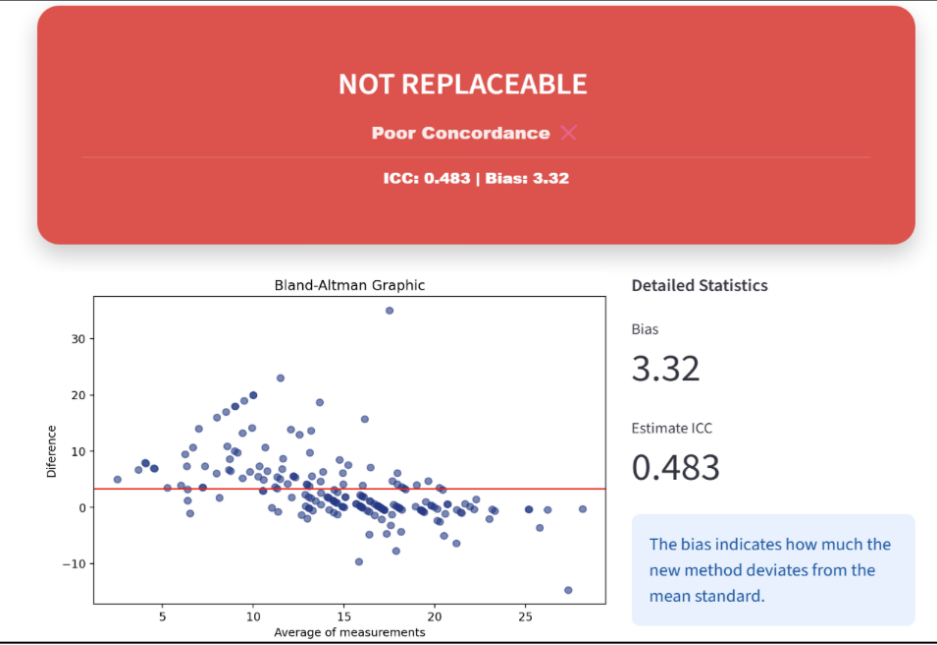


Figure A.15: Mockup Results